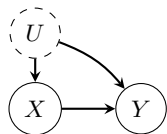


Structure-specific
(correct)

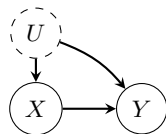


True: Conf-Desc

Apply $R_{\mathcal{G}_2}$:
 $P(Y|\phi(X))$ invariant

Good OOD
Test error: low

Structure-specific
(wrong)



True: Conf-Desc

Apply $R_{\mathcal{G}_1}$:
 $\phi(X) \perp E \mid Y$

Poor OOD
Test error: high

Structure-agnostic
(SaCRL, ours)

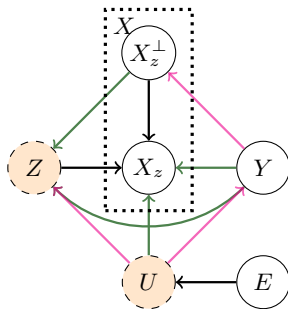


Unknown structure

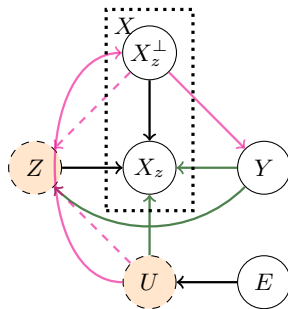
Learn α :
auto-select $R_{\mathcal{G}_k}$

Good OOD
Robust to structure

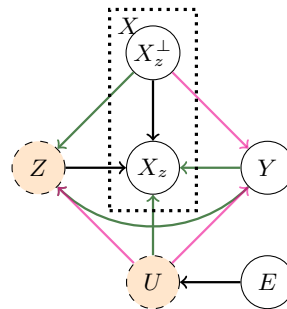
Candidate causal structures



(a) $\mathcal{G}_1$: anti-causal



(b) $\mathcal{G}_2$: confounded-descendant



(c) $\mathcal{G}_3$: confounded-outcome